Flow-Aware Latent Diffusion Model for Temporal Cerebral DSA Synthesis

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Abstract. Digital Subtraction Angiography (DSA) is a cornerstone modality for vascular diagnosis and intervention, providing visualization of contrast agent dynamics over time. However, large-scale temporal DSA data are difficult to obtain due to acquisition cost, radiation exposure, and strict clinical data privacy constraints that limit data sharing. These challenges motivate realistic temporal DSA synthesis for data augmentation and simulation, yet make the task technically demanding. In this work, we present the first framework for fully synthetic temporal cerebral DSA generation. Our approach is built upon a flow-aware diffusion model in the latent space of an autoencoder and introduces a content–motion factorization that disentangles static vascular anatomy from dynamic contrast propagation. The dynamic component is explicitly realized via a flow-based motion mechanism, providing a strong inductive bias for coherent contrast evolution while preserving anatomical structure. We evaluate the proposed method on real clinical DSA data using Fréchet Video Distance and clincal expert assessment via Likert-scale ratings, demonstrating clinically plausible and temporally coherent DSA sequence synthesis. Our code is available at **github.com/philo1123/miccai-ldm-anon**.

Keywords: Diffusion model · DSA · Image generation.

1 Introduction

Digital Subtraction Angiography (DSA) is a cornerstone imaging modality for vascular diagnosis and interventional procedures [3,17], providing high-resolution visualization of contrast agent propagation within blood vessels. Beyond static imaging, temporal DSA sequences capture dynamic information such as contrast agent inflow, perfusion, and washout, which is critical for functional assessment of vascular conditions. However, the acquisition of high-quality temporal DSA data is costly and associated with radiation exposure, resulting in limited data availability. These challenges motivate the development of generative models capable of synthesizing high-fidelity DSA data to support data augmentation, simulation, and downstream learning tasks [11,20].

Related Work and Problem Statement Prior work related to generative modeling in DSA and X-ray angiography has predominantly addressed task-specific or conditional synthesis problems, rather than large-scale image generation as a standalone objective. Representative studies include generating pseudo-angiograms to facilitate downstream segmentation or registration [8,14,23], translating between native angiograms and alternative angiographic views [5,12], and predicting future frames conditioned on previously observed sequences [1,11]. While valuable within their respective scopes, these methods are largely input-conditioned (e.g., translation or future-frame prediction) and therefore do not address the synthesis of diverse, high-fidelity DSA sequences in a fully synthetic setting without requiring an observed sequence as input. More recently, diffusion-based generation has begun to be explored for single-frame DSA synthesis [21], highlighting the growing interest in fully synthetic angiographic image generation while leaving temporal modeling largely unaddressed.

Recent advances in diffusion models have demonstrated strong performance in medical image synthesis due to their stable training dynamics and high sample fidelity [9,16,19]. However, directly extending existing image or video diffusion frameworks to temporal DSA generation is non-trivial. Unlike natural videos, where both appearance and geometry often change over time, DSA sequences exhibit largely static vascular anatomy, while temporal variation is dominated by structured contrast agent propagation along vessel trees. This observation aligns with recent video diffusion studies that emphasize the importance of disentangling content and motion representations for improved temporal consistency [24], but existing approaches primarily rely on token-based transformer architectures [15,24] developed for semantic-rich natural videos, which may provide weak inductive bias for contrast-driven angiographic sequences.

In contrast, flow-style warping offers a simple mechanism to impose explicit motion structure in generative models by predicting a dense deformation field and synthesizing frames via differentiable resampling [10]. Related ideas have been explored for video frame synthesis using learned voxel/flow fields and warping-based generation [13]. In medical imaging, deformation fields are a standard representation for non-rigid alignment, and learning-based registration methods further motivate the use of deformable warping modules for modeling structured temporal variation [2]. This perspective is particularly relevant to temporal DSA, where vascular anatomy is largely stationary and temporal changes are dominated by the hemodynamic progression of the contrast agent.

Our Contributions In this work, we develope a fully synthetic temporal DSA generation and introduce a **flow-aware latent diffusion framework** tailored to contrast agent-driven angiographic dynamics (Fig. 1). Our key contributions are threefold: (i) a content–motion factorization (CMF) that conditions generation on a static vascular content frame while modeling only the dynamic component; (ii) an explicit flow-based motion realization (FMR) that synthesizes sequences via differentiable warping, providing a strong inductive bias for contrast propagation; and (iii) we adopt a temporal transformer within a U-Net denoiser to encourage long-range consistency across frames [7]. Extensive exper-

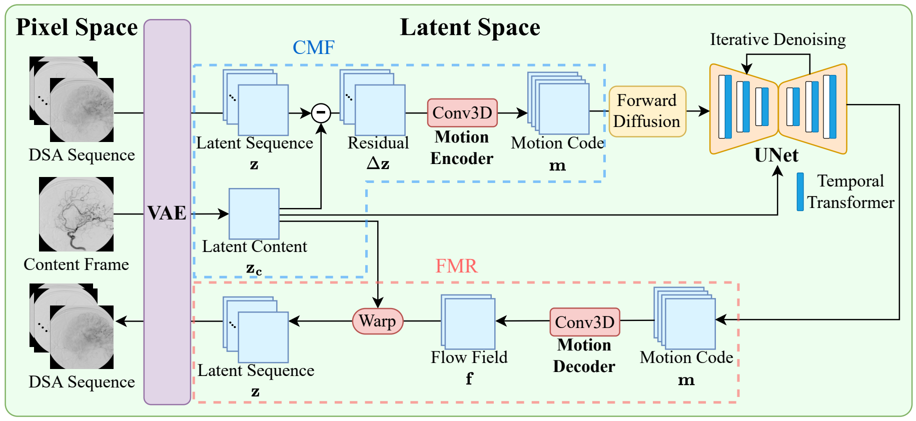

Fig. 1. Architecture of our flow-aware latent diffusion model for temporal DSA generation. CMF decomposes the VAE latent sequence into a content latent $\mathbf{z}_c$ and a motion code $\mathbf{m}$; a diffusion U-Net denoiser with a temporal transformer generates $\mathbf{m}$ conditioned on $\mathbf{z}_c$. FMR converts $\mathbf{m}$ to a flow field $\mathbf{f}$ and warps $\mathbf{z}_c$ to produce the latent sequence, which is decoded to DSA frames by the VAE.

iments, including distributional and paired metrics as well as expert assessment, demonstrate improved realism and temporal coherence for both real-content conditioned and fully synthetic generation.

2 Flow-Aware Latent Diffusion Framework

We assume a dataset with grayscale DSA sequences and let $\mathbf{x} \in \mathbb{R}^{B\times 1\times T\times H\times W}$ be a data batch of size B with T number of frames and image resolution (H, W). Inspired by the latent diffusion model (LDM) [16], we first encode $\mathbf{x}$ into latents $\mathbf{z} = \mathcal{E}(\mathbf{x}) \in \mathbb{R}^{B\times C\times T\times h\times w}$ using a variational autoencoder (VAE) applied frame-wise, where $\mathcal{E}$ and $\mathcal{D}$ denote the VAE encoder and decoder, respectively, (h, w) the latent resolution, and C the latent channel dimension. Since vascular anatomy is largely static while temporal variation is dominated by contrast agent propagation, we then factorize $\mathbf{z}$ into a latent content $\mathbf{z}_c$ (static anatomy) and a compact motion code $\mathbf{m}$ (dynamic contrast), referred to as **Content-Motion Factorization (CMF)**. A content-conditioned diffusion model generates motion codes, which are subsequently transformed into latent sequences via **Flow-based Motion Realization (FMR)**. At inference time, our pipeline synthesizes a full sequence from content conditioning in an end-to-end, differentiable manner. We describe these components in detail below.

2.1 Content–Motion Factorization and Motion Diffusion (CMF)

We model temporal DSA dynamics in the VAE latent space by factorizing each sequence into a static latent content and a compact motion code. This design encourages anatomical consistency across frames and allows the diffusion model to focus on contrast-driven temporal evolution.

Latent Factorization Given a latent sequence $\mathbf{z} \in \mathbb{R}^{B\times C\times T\times h\times w}$ encoded from $\mathbf{x}$, we extract a representative latent content $\mathbf{z}_c \in \mathbb{R}^{B\times C\times 1\times h\times w}$ and replicate it along time T. We then define residual dynamics as $\Delta\mathbf{z} = \mathbf{z} - \mathrm{repeat}(\mathbf{z}_c)$. A shallow 3D convolutional motion encoder $\mathcal{M}_{\mathrm{enc}}$ compresses $\Delta\mathbf{z}$ into a compact motion code $\mathbf{m} = \mathcal{M}_{\mathrm{enc}}(\Delta\mathbf{z}) \in \mathbb{R}^{B\times P\times T\times \tilde{h}\times \tilde{w}}$, which serves as the target of our motion diffusion model, where P denotes the motion-code channel dimension and $(\tilde{h}, \tilde{w})$ the spatial resolution of the motion code.

Content-Conditioned Motion Diffusion We learn the conditional distribution of motion codes $p_\theta(\mathbf{m} \mid \mathbf{z}_c)$ using a standard diffusion process. Let $\mathbf{m}_t$ denote the noisy motion code at diffusion step t and $\boldsymbol{\epsilon}_\theta$ be a U-Net denoiser that predicts the added noise. We condition the denoiser on $\mathbf{z}_c$ by channel-wise concatenation at the U-Net input, yielding a simple and effective anatomical guidance mechanism. The model is trained using a noise-prediction objective

$$\mathcal{L}_{\mathrm{diff}} = \mathbb{E}_{t,\boldsymbol{\epsilon}} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{m}_t, t \mid \mathbf{z}_c) \right\|_2^2 . \tag{1}$$

During training, the latent content $\mathbf{z}_c$ is extracted from each real DSA sequence by selecting a mid-sequence frame: we uniformly sample one of the two central frames, which typically exhibits clearer vessel opacification and provides stable anatomical conditioning. At inference time, $\mathbf{z}_c$ can be obtained either from a real DSA frame or from a pretrained single-frame DSA generator [21], enabling unconditional synthesis of temporal DSA sequences from scratch.

Temporal Transformer in the Denoiser Since $\mathbf{m}$ represents a temporal sequence, we augment the U-Net with a temporal transformer module adapted from AnimateDiff [7] to couple information across the frame dimension T. This module is inserted into U-Net blocks that already employ attention (typically at lower spatial resolutions) to capture long-range temporal dependencies while retaining convolutional spatial inductive bias with limited overhead.

2.2 Flow-based Motion Realization (FMR)

Given a motion code $\mathbf{m} \in \mathbb{R}^{B\times P\times T\times \tilde{h}\times \tilde{w}}$ sampled from the content-conditioned diffusion model (Sec. 2.1), we convert it into an explicit motion field and synthesize the full latent sequence by warping a static latent content as described in the following.

Motion-to-Flow Decoding We first decode $\mathbf{m}$ into a dense latent-space flow field $\mathbf{f} = \mathcal{M}_{\mathrm{dec}}(\mathbf{m}) \in \mathbb{R}^{B\times 2\times T\times h\times w}$, with a 3D convolutional motion decoder

$\mathcal{M}_{\text{dec}}$, where the two channels correspond to horizontal and vertical displacements. To stabilize training and constrain motion magnitude, we apply a bounded re-parameterization per frame:

$$\tilde{\mathbf{f}}_t = \alpha \tanh(\mathbf{f}_t), \quad t = 1, \ldots, T, \tag{2}$$

where $\alpha > 0$ controls the maximum displacement amplitude in latent-space.

Warping-Based Sequence Synthesis For each time step t, we then synthesize the latent frame $\hat{\mathbf{z}}_t$ by sampling $\mathbf{z}_c$ at displaced coordinates determined by $\tilde{\mathbf{f}}_t$:

$$\hat{\mathbf{z}}_t(i, j) = \mathbf{z}_c(i + \tilde{v}_t(i, j),\, j + \tilde{u}_t(i, j))\,, \tag{3}$$

where $\tilde{\mathbf{f}}_t = (\tilde{u}_t, \tilde{v}_t)$ and (i, j) index spatial locations on the $h \times w$ latent grid. Since the displaced coordinates are generally non-integer, we use standard bilinear interpolation as in spatial transformer networks [10]. Stacking all frames yields $\hat{\mathbf{z}} \in \mathbb{R}^{B \times C \times T \times h \times w}$, and the final sequence is obtained by decoding each frame with the VAE decoder $\mathcal{D}$, i.e., $\hat{\mathbf{x}} = \mathcal{D}(\hat{\mathbf{z}})$.

3 Experiments and Results

3.1 Experimental Setup

We trained and evaluated our model on an anonymized clinical DSA dataset with 9,642 sequences. Each sequence was preprocessed into T=8 frames at 256×256 resolution, yielding 77,136 frames in total.

We used a VAE with a latent size of $3 \times 64 \times 64$, and represented motion with a compact code of size $8 \times 32 \times 32$. The flow scaling factor in Eq. 2 is set to $\alpha = 3$. First, we trained the VAE and then trained the diffusion model on frozen VAE latents. The VAE was trained with AdamW (learning rate $2 \cdot 10^{-5}$), $B = 2$ batches of sequence length $T = 8$, while the diffusion model was trained with 1,000 diffusion steps using AdamW (learning rate $5 \cdot 10^{-5}$, batch size of 12) and sampled with deterministic DDIM sampling [18] at inference. All models were trained on a single NVIDIA A100 80 Gb GPU.

3.2 Results

We consider two generation settings: **Real-Content Conditioned (RCC)** that conditions on a content frame extracted from each real sequence, enabling paired evaluation; and **Synthetic-Content Conditioned (SCC)** that is conditioning on a content frame sampled from a pretrained single-frame DSA generator [21], yielding fully synthetic (unpaired) sequences. We evaluate our method using quantitative metrics for distributional quality, frame-level fidelity (when paired), and temporal consistency. Additionally, we conduct a blinded clinical expert study to rate generated DSA sequences on a 1–10 Likert realism scale.

Table 1. Quantitative results under RCC and SCC. "–" indicates not applicable for SCC due to the unpaired setting.

Setting	FVD (↓)	SSIM (↑)	PSNR (↑)	LPIPS (↓)	tLPIPS (↓)
RCC	26.77	0.894	28.19	0.131	0.0166
SCC	81.02	–	–	–	0.0210

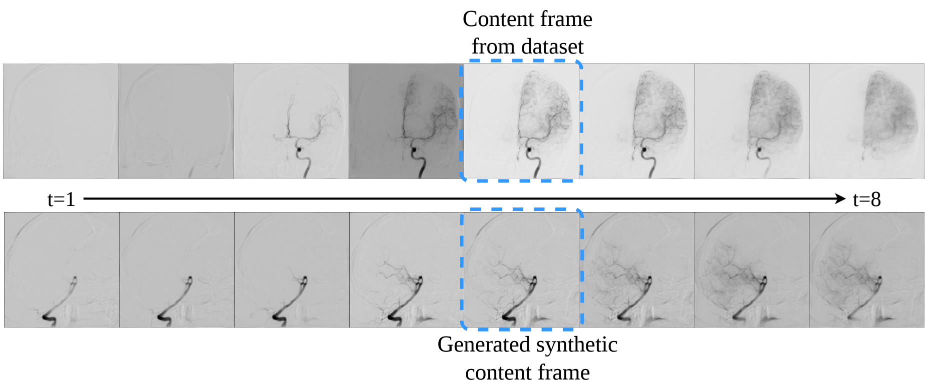

Fig. 2. Examples under RCC (top) and SCC (bottom); the dashed box marks the content frame used for conditioning (real dataset frame for RCC, synthetic frame for SCC) to generate the temporal sequence. Left-to-right: $t = 1 \to t = 8$.

Quantitative Results We report Fréchet Video Distance (FVD) for the distribution-level comparison and SSIM/PSNR/LPIPS under the RCC setting. To quantify temporal consistency in both RCC and SCC, we additionally use tLPIPS [4,22] at the statistics-level, which measures the perceptual change between consecutive frames. For a sequence $\mathbf{x}$ of length T with t-th frame x_t and generated sequence $\hat{\mathbf{x}}$, we define the intra-sequence temporal change as $\ell(\mathbf{x}) = \frac{1}{T-1} \sum_{t=2}^{T} \mathrm{LPIPS}(x_t, x_{t-1})$, and then report $\mathrm{tLPIPS} = |\mathbb{E}[\ell(\hat{\mathbf{x}})] - \mathbb{E}[\ell(\mathbf{x})]|$.

Table 1 demonstrates that our framework produces high-fidelity temporal DSA sequences under both the RCC and the more challenging fully synthetic SCC setting. Under RCC, where generations are content-paired with real sequences, we achieve a low FVD of 26.77 together with strong frame-level fidelity (SSIM 0.894 / PSNR 28.19 / LPIPS 0.131), indicating that the proposed content-conditioned motion modeling preserves vascular anatomy while synthesizing plausible contrast propagation. In the fully unpaired SCC setting, FVD increases to 81.02, likely driven by the unavoidable content distribution gap and the known content sensitivity of FVD [6]. Crucially, despite being fully synthetic, temporal behavior remains close to real data: the statistics-level tLPIPS stays small (0.0210 in SCC vs. 0.0166 in RCC), suggesting that the generated sequences reproduce contrast-driven temporal changes rather than merely matching static appearance. Fig. 2 further provides representative examples showing coherent contrast evolution across frames.

Table 2. Expert Likert ratings (1–10) aggregated at the sample level (mean across 3 blinded doctors).

Setting	Mean ± Std	Median [Q1, Q3]
Real	7.91 ± 1.43	8.17 [7.0, 9.0]
RCC	6.00 ± 2.28	6.33 [4.1, 8.0]
SCC	5.77 ± 2.00	5.83 [4.1, 7.6]

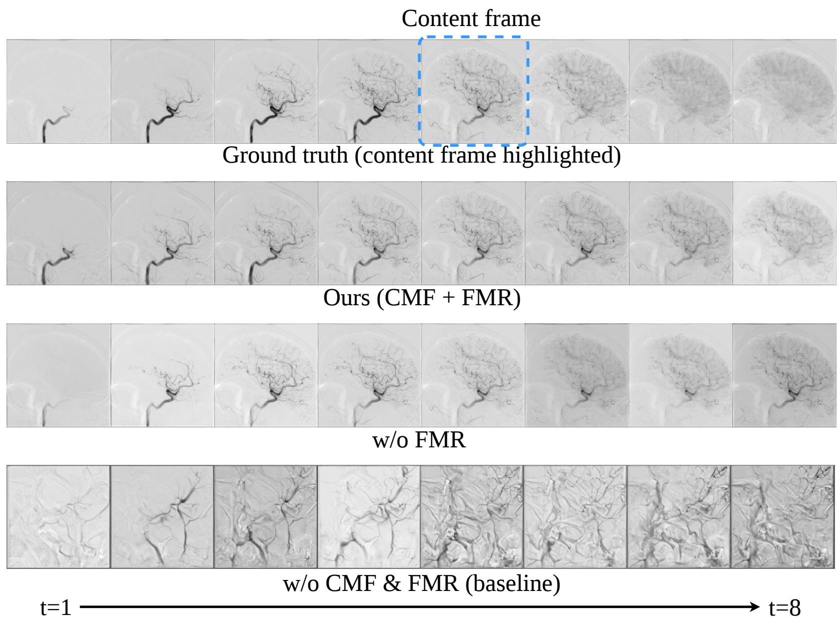

Fig. 3. Ablation of CMF and FMR under the RCC setting. The bottom row w/o CMF & FMR corresponds to the base temporally-augmented LDM (LDM+temporal transformer). Left-to-right: $t = 1 \rightarrow t = 8$.

Expert Evaluation We further assess clinical realism via a blinded expert study with three domain experts (2 neuroradiologists, 1 neurosurgeon). We sampled 150 sequences in total, including 50 real, 50 RCC, and 50 SCC samples. Sequences were presented in random order and the raters were blinded to the source of each sample. Each sequence received a single overall realism score on a 1–10 Likert scale with score 1 being unrealistic and 10 true/realistic DSA.

Table 2 summarizes the blinded expert evaluation. Real sequences achieved the highest realism scores, followed by RCC and SCC, establishing a consistent ordering (Real > RCC > SCC). RCC is preferred over SCC, indicating that conditioning on real anatomical content improves perceived realism. Importantly, even in the fully synthetic SCC setting, sequences were rated in a moderate realism range rather than collapsing to implausible appearance, supporting the effectiveness of our motion modeling framework under blinded assessment.

Table 3. Quantitative ablation under RCC.

Method	FVD (↓)	SSIM (↑)	PSNR (↑)	LPIPS (↓)	tLPIPS (↓)
Ours (CMF+FMR)	26.77	0.894	28.19	0.131	0.0166
w/o FMR	98.08	0.881	26.57	0.141	0.0297

Ablation Study To isolate the contribution of our two key components, CMF and FMR, we evaluate two ablated variants under the RCC setting (Fig. 3). The **w/o CMF & FMR** variant serves as our baseline, directly extending latent diffusion to temporal synthesis via a temporally-augmented LDM (LDM+temporal transformer). This baseline follows AnimateDiff [7]: a temporal U-Net denoiser that directly generates the full latent sequence without CMF or FMR. The variant **w/o FMR** retains CMF but removes flow-based warping; instead, it decodes motion as a residual latent sequence $\widehat{\Delta \mathbf{z}}$ and synthesizes frames by $\hat{\mathbf{z}}_t = \mathbf{z}_c + \widehat{\Delta \mathbf{z}}_t$.

As shown in Fig. 3, the **w/o CMF & FMR** baseline (i.e., LDM+temporal transformer) collapses visually, making quantitative metrics uninformative; we therefore omit scores for this variant. When removing only FMR, the synthesized sequence exhibits weaker temporal variation and reduced frame-to-frame coherence compared to the full model, consistent with the quantitative degradation in Table 3. We additionally conduct a blinded human ablation study under both RCC and SCC with a *paired* design: for each case, the full model and **w/o FMR** variant are conditioned on the *same* content frame (real in RCC; synthetic in SCC). Using 25 paired sequences per setting, adding FMR improved expert ratings by $+1.52 \pm 1.83$ (RCC) and $+1.64 \pm 1.75$ (SCC) in mean paired difference±SD. These results further support that FMR enhances clinically perceived realism by enforcing coherent contrast-driven motion.

Overall, these results demonstrate that CMF stabilizes anatomy-conditioned temporal generation, while FMR is critical for realizing coherent motion patterns via explicit flow-based warping.

4 Conclusion

To our knowledge, we present the first framework for fully synthetic temporal cerebral DSA generation, explicitly modeling contrast-driven dynamics beyond input-conditioned translation or future-frame prediction. By factorizing content and motion and realizing temporal evolution through flow-based warping in latent space, our approach achieves high-fidelity vascular appearance and coherent contrast propagation. Extensive quantitative evaluation and blinded expert assessment demonstrate the effectiveness of the proposed design under both real-content conditioned and fully synthetic settings. Future work will investigate leveraging the proposed generator as a synthetic data engine to mitigate the limited availability of large-scale temporal DSA data and to support training downstream models such as representation pretraining for vessel analysis and temporal frame interpolation/sequence densification for low-dose or low-frame-rate DSA protocols.

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